CAN RETRIEVED EDITING EXPERIENCE TRANSFER ACROSS NEURAL CHECKPOINTS? A CONTROLLED TINY-NETWORK STUDY

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ABSTRACT

Previously observed model edits may help a language-model editor repair new neural checkpoints, but examples can also transfer poorly. We evaluate a frozen library of successful and unsuccessful restricted weight edits on independently initialized tiny classifiers. A fixed editor receives either no experience, two randomly selected records, or two records selected using task and weight statistics. On twelve held-out confirmation checkpoints, the mean final-accuracy differences for relevant retrieval versus no experience and random examples are +0.114 and +0.024 percentage points, respectively; the intersection–union p-value is 0.5 and only one target favors retrieval over both controls. The results do not meet a prespecified five-point threshold. These observations apply to a small synthetic benchmark and a single editor, and do not establish that weight-editing experience cannot transfer in other settings.

1 INTRODUCTION

Weight-space programming explores whether language models can interpret and modify model parameters directly (Takehi et al., 2026). When a model produces an edit, the resulting program and its failures could potentially inform later editing attempts. However, past examples may help merely by demonstrating the output format, and an edit that helped one checkpoint may harm another. We study whether retrieval based on task and weight characteristics adds value beyond randomly selected examples under the same editing interface.

The question relates to executable skill libraries, which have been used for agent behavior (Wang et al., 2023), and to transfer of task vectors between distinct pretrained models (Kawamoto et al., 2025). Task arithmetic combines differences between pretrained and fine-tuned weights (Ilharco et al., 2023). Separately, SERAC stores factual edits in external memory and uses retrieval to condition predictions while retaining the base weights (Mitchell et al., 2022). Our focus is narrower: reuse of restricted sparse editing experience across independently trained tiny classifiers, including unsuccessful experiences and potential negative transfer. It does not establish novelty for retrieval or parameter transfer in general.

2 EXPERIMENTAL DESIGN

Each target is a two-layer $2 \to 16 \to 2$ ReLU classifier trained on a synthetic two-dimensional classification task. Training is completed before any edit; the editing interface exposes no target gradients. We construct archive, development, screen, and confirmation namespaces with independent seeds. Confirmation includes held-out task-parameter combinations. Eligibility is determined from baseline acceptance-set accuracy; all screened-out candidates remain recorded.

Each checkpoint is initialized with an independent seed and trained for 12 full-batch SGD steps at learning rate 0.12 on 128 samples. The three binary task families use, respectively, the rules $x_0 + px_1 > 0$, $x_0^2 + x_1^2 > p$, and $(x_0 > p)$ xor $(x_1 > 0)$. Inputs are sampled from $[-1, 1]^2$. Training uses 75% samples from $x_0 \in [-0.4, 1]$ and 25% from $[-1, -0.4]$; support, acceptance and final data use the first region, while guard data use the second. Per target there are 24 support, 128

acceptance, 128 guard and 512 final examples, drawn separately. Candidates are retained only if baseline acceptance accuracy lies in $[0.55, 0.90]$; the first eligible checkpoints are selected in seed order. The archive attempt used 12 such checkpoints, of which four yielded nonempty edits with strictly positive acceptance gain and an acceptable guard change. The remaining eight records are retained in the library and may be retrieved.

An editor can specify at most eight scalar additions. Each addition uses a coefficient from a fixed discrete set multiplied by the original tensor's root-mean-square magnitude, with a fixed floor. The permitted coefficients are $\{-2, -1, -0.5, 0.5, 1, 2\}$ and the RMS floor is 0.01. The editor receives the same target weights and labeled support examples across conditions. A candidate is accepted only if acceptance accuracy does not decline and regional guard accuracy falls by at most ten percentage points. Guard examples cover a distinct input region of the same task, not a separate retained capability; 25% of initial training examples come from this region.

We freeze a source archive before screen. Each source record contains checkpoint identity, task description, patch, validation outcome, and the realized parameter difference; failures remain in the archive. A fixed score combines task-description TF–IDF cosine similarity with normalized tensor RMS proximity at weights 0.8 and 0.2, respectively, and selects two architecture-compatible records in the relevant condition. The weights were fixed a priori; the four generated development targets were not used for empirical tuning. The random condition receives two compatible records selected by a fixed seed, with matching fields and budgets. The no-experience condition receives no example. All conditions use the same editor and proposal budget. The requested endpoint model was `kiro/claude-haiku-4.5`, which returned identifier `claude-haiku-4.5`; temperature was 0.7, output was capped at 1,024 tokens, and calls were serialized. The provider did not expose an immutable revision. One validator-error correction is allowed and both requests are recorded; infrastructure and parsing failures remain in the denominator.

Final examples are separately stored and are not supplied to the editor. For accepted proposals, a separate evaluator applies the edit to the original checkpoint; invalid and rejected proposals retain the unedited baseline in the intention-to-treat analysis. We average repeats within each target and compare relevant retrieval against both controls. The intersection–union statistic is the larger of the two one-sided paired p-values. We also report per-target outcomes and uncertainty intervals, treating the checkpoint as the independent unit. The paired sign-flip test assumes exchangeability of signs under the null; small numbers of checkpoints limit its interpretation. The prespecified practical threshold is a five-percentage-point mean gain over each control, with a majority of targets improving against both and no material additional guard damage. Mechanism controls compare a source raw difference, re-execution of its RMS-scaled sparse program, and an editor-generated proposal; their source records may differ and are not assumed to be source-matched.

3 RESULTS

The exploratory screen includes eight independently initialized eligible targets, with two proposals per arm and target. After the prespecified acceptance and regional guard gate, relevant retrieval exceeds no experience by 0.48 percentage points on average (95% paired bootstrap interval [0.00, 1.43] percentage points) and random retrieval by 0.72 points ([0.00, 2.16]). The one-sided paired sign-flip p-value is 0.5 for each contrast, and therefore the intersection–union p-value is 0.5. The 99% intervals are [0.00, 1.90] and [0.00, 2.88] points, respectively; nominal Bonferroni-adjusted 97.5% intervals are [0.00, 1.43] and [0.00, 2.16]. Only one of eight targets favors relevant retrieval over both controls. These exploratory results do not meet the prespecified five-point threshold or the majority-target condition. Invalid and rejected proposals are retained as zero-gain outcomes. Among 16 proposals per arm, one no-experience proposal was invalid; accepted counts were 5, 4, and 8 for no experience, random, and relevant, respectively. Mean guard changes under the intention-to-treat rule were +0.29, +0.39, and +0.78 percentage points, respectively. Independent replay has checked the raw responses, selections, target-level contrasts, intervals, and one example of each mechanism control.

On the separately held-out confirmation set of 12 checkpoints (24 proposals per arm), relevant retrieval exceeds no experience by 0.114 percentage points (95% interval [0.000, 0.293]) and random retrieval by 0.024 points (95% interval $[-0.114, 0.203]$). The 99% intervals are [0.000, 0.382] and $[-0.130, 0.277]$ points, respectively; nominal Bonferroni-adjusted 97.5% intervals are [0.000,

Table 1: Mean final accuracy and gain over the unedited baseline, in percentage points. Each row averages two proposals per target; rejected or invalid proposals retain the baseline.

Split	Condition	Targets	Final accuracy (%)	Gain (pp)
Screen	None	8	71.863	0.964
Screen	Random	8	71.619	0.720
Screen	Relevant	8	72.339	1.440
Confirm	None	12	70.972	0.106
Confirm	Random	12	71.061	0.195
Confirm	Relevant	12	71.086	0.220

0.358] and $[-0.130, 0.220]$. The corresponding one-sided sign-flip p-values are 0.25 and 0.5, yielding an intersection–union p-value of 0.5. One of 12 targets favors relevant retrieval over both controls. These findings do not meet the five-point or majority-target criteria. All 72 confirmation proposals were valid after any permitted correction, and 8, 9 and 11 were accepted in the no-experience, random and relevant arms, respectively. The corresponding mean guard changes were +1.14, +1.89 and +2.28 percentage points under the intention-to-treat rule.

For the exploratory confirmation mechanism comparison, direct raw-difference transfer, RMS-scaled program execution, and relevant editor regeneration yielded mean final gains of 1.63, 0.75, and 0.22 percentage points, respectively; raw and program edits passed the same acceptance gate on four of twelve targets each. These are descriptive results, not source-matched causal comparisons: the highest-ranked validated source used by the two transport controls need not be among the two records given to the editor. Percentile intervals with few targets and many tied outcomes have limited resolution. The sign-flip test assumes symmetry, and an apparently nonnegative bootstrap interval endpoint is not evidence that harm is impossible.

4 Limitations

The targets are small classifiers with one architecture, and the guard set tests a held-out region of the same task. The small number of independently initialized checkpoints and only three task families limit statistical precision and diversity. Repeats for the same checkpoint are not counted as independent targets. The 95% and 99% percentile bootstrap intervals are approximate and can be unstable with sparse nonzero effects; a nonnegative interval endpoint does not rule out harm in a broader population. These experiments cannot establish transfer to language models or substantially different architectures. The confirmation implementation was frozen after screen proposals began but before screen final labels were opened; during screen collection we reopened two related papers and the WSP blog already cited in the original research brief, contrary to its prohibition on further online literature checks during testing. The retrieved material did not enter the frozen archive, editor prompts, or screen method; acceptance and guard outcomes could have been visible by that time. Provider model revision identifiers may not be available; request and returned model identities, prompts, and responses must therefore be reported alongside this limitation.

5 Reproducibility and LLM Use

This paper is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0). The experiment records configuration, checkpoint and data hashes, prompts, responses, validation outcomes, failed attempts, and independent checks. Across archive, screen, and confirm, the provider recorded 155 requests, including validator-correction requests, approximately 1.016 million input and 0.033 million output tokens. Two preliminary connectivity requests are additional, one of which failed and has unknown usage. Monetary cost was not exposed and is unknown; the model calls used an existing account. All target computation used one CPU thread and no GPU. Frozen source programs perform fixed-coordinate, RMS-scaled additions; they are a restricted form of adaptation and are not evidence of general program portability. The screen had eight checkpoints and the confirm set has twelve, with two editor repeats per condition and checkpoint. AI assistants were used to implement and review the experimental workflow and to prepare this manuscript; the author di-

rected the study and confirmed the authorship and affiliation; independent Codex subagents checked implementation and results.

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